DATA 266 Homework 2

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Abstract

In many real-world applications, Large Language Models (LLMs) are used to reason over structured tabular data rather than plain text. Examples include risk assessment, customer churn prediction, and medical triage systems. We investigate whether advanced prompt engineering strategies improve LLM performance when making predictions from structured data. We use LLMs for a tabular reasoning task, design and compare prompt engineering techniques, evaluate predictions using classification metrics, and analyze where LLM reasoning fails.

Index Terms

Data processing, LLMs

I. Prompting Techniques

We have documented all prompt techniques used:

- System-prompt: Sample available Appendix C
- Zero-shot: Sample available Appendix D
- 5-shot: Sample available Appendix E
- Chain-of-Thought (CoT): Sample available Appendix F
- Self-consistency: Sample available Appendix G
- Tree-of-Thought: Sample available Appendix H

II. Prompt Engineering for Tabular Prediction using LLMs

TABLE I
Results Table

LLM	Prompting Method	Accuracy	Precision	Recall	F1
google/gemini-3-flash-preview	Zero-shot	**0.990**	0.994709	**0.979167**	**0.986877**
google/gemini-3-flash-preview	Five-shot	0.988	**1.000000**	0.968750	0.984127
google/gemini-3-flash-preview	CoT	0.982	0.989305	0.963542	0.976253
google/gemini-3-flash-preview	Self-consistency (3)	0.986	**1.000000**	0.963542	0.981432
google/gemini-3-flash-preview	ToT	0.974	0.994475	0.937500	0.965147
openai/gpt-4.1-nano	Zero-shot	0.780	0.879630	0.494792	0.633333
openai/gpt-4.1-nano	Five-shot	0.776	0.785714	0.572917	0.662651
openai/gpt-4.1-nano	CoT	0.692	0.588785	0.656250	0.620690
openai/gpt-4.1-nano	Self-consistency (3)	0.758	0.669856	0.729167	0.698254
openai/gpt-4.1-nano	ToT	0.616	0.000000	0.000000	0.000000

Full comparison results table comparing all prompting techniques.

A. Hard Case Analysis

1) gemini-3-flash-preview: Full reasoning traces for 9 missclassified samples for Gemini 3 Flash are at Appendix A.

The model performed slightly worse when prompted for greater reasoning, overthinking on our system information. The model primarily missed 3rd-class males of whom survived, passengers of ID 82, 205, 268, 745 and 401. Passenger 808, an 18-year-old 3rd class female, was predicted to have survived, but did not, which was a statistical anomaly. The model primarily made these misclassifications due to particular statistical assumption priors. However, the mode explicitly hallucinates passenger ID 70 as a "documented historical anomaly" whom survived, though this is false.

The Gemini model is exceptionally more accurate, faster and at higher availability than other SOTA models. It's likely it was pre-trained on this particular problem. The next highest achiever was openai/gpt-5.2-chat (missed logging it) at 85% accuracy using zero-shot. Because of this, I believe Gemini likely has easily-accessible parametric knowledge about this particular dataset, and that our particular prompts were able to elicit the particular pre-training configuration it was trained on.

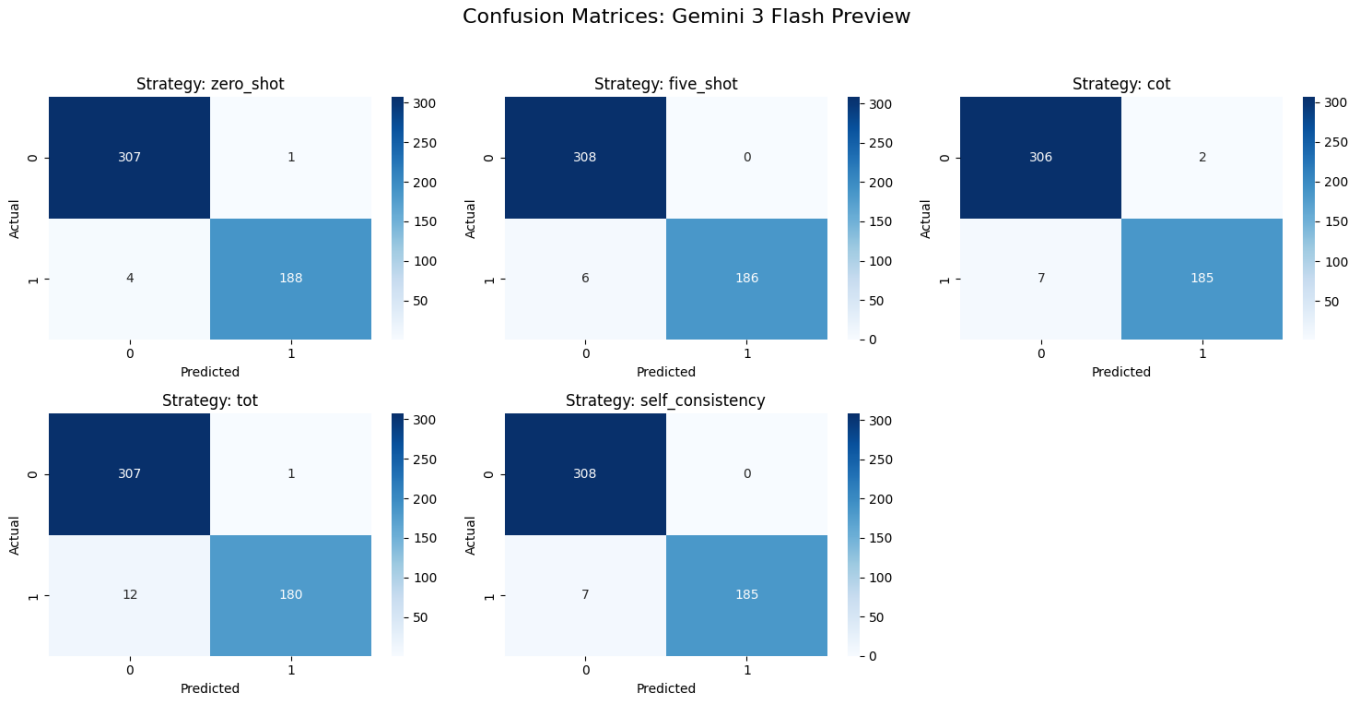

Fig. 1. Confusion Matrices for Gemini Flash 3 Preview on the Kaggle Titanic dataset

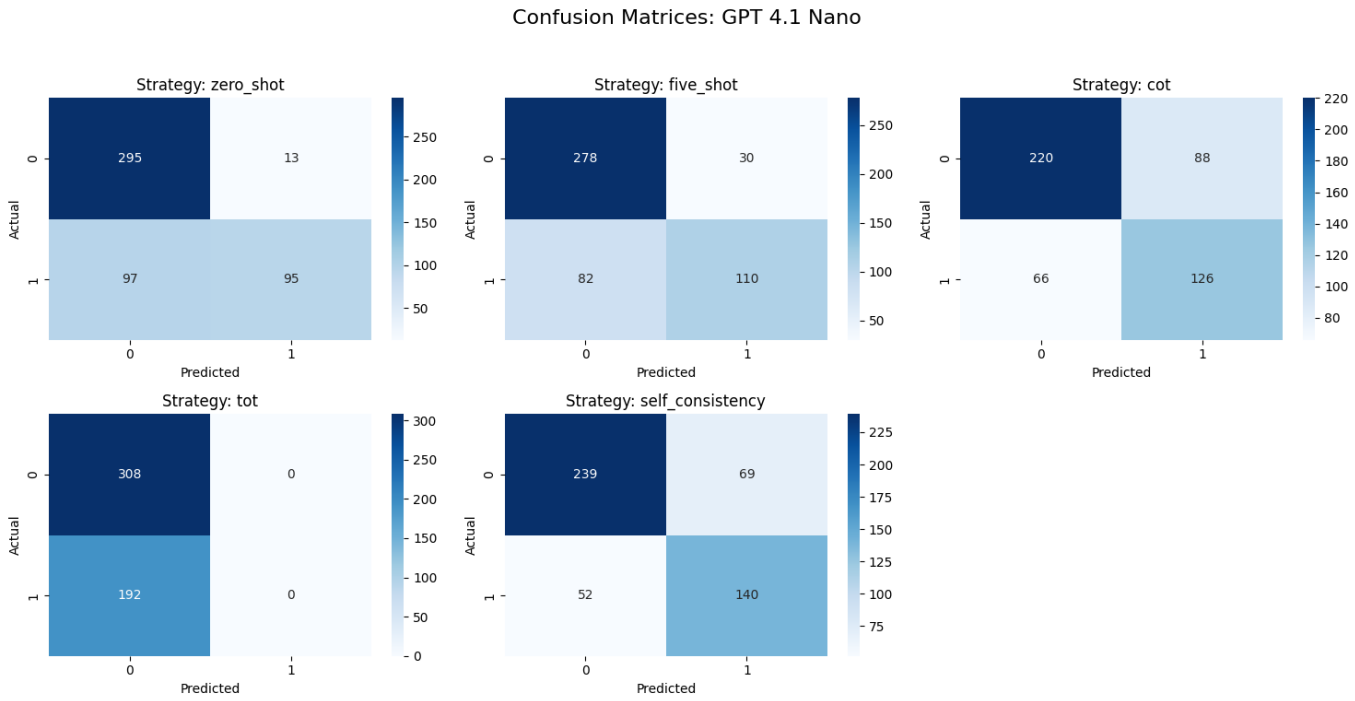

Fig. 2. Confusion Matrices for GPT 4.1 Nano on the Kaggle Titanic dataset

2) gpt-4.1-nano: Full reasoning traces for 9 missclassified samples for GPT 4.1 Nano at Appendix B.

Our system prompt elicited the model to consider anomalies in survival, and thus some reasoning responses directly contradicted known statistical knowledge about the event. For example, for passengers 55 and 140, higher-class males were predicted to have survived, despite them having greater statistical likelihood of dying. The model also came to underestimate 3rd-class women traveling alone. Overall, enhanced reasoning degraded model performance, where the shorter prompts likely engaged activations with more dependence on direct statistical assumptions.

III. Enterprise Retrieval-Augmented Generation (RAG) with WixQA

A. Baseline RAG Implementation

TABLE II
BASELINE RAG SYSTEM CONFIGURATION

	Configuration
Embedding model	all-MiniLM-L6-v2
Vector database	PgVector
Retriever type	HNSW
Generator model	gpt-oss-120b
Prompting Strategy	Zero-shot

Baseline system configuration for our baseline RAG implementation.

Our baseline configuration uses lightweight

TABLE III
WIXQA-SYNTHETIC 750-SUBSET BASELINE RESULTS

Chunk Size	Overlap	k	Context Recall	F1	ROUGE-1	Observation
500	60	1	0.4790	0.35658	0.44194	
500	60	3	0.6560	0.3574	0.4512	
500	60	5	0.7348	0.35291	0.45046	
500	60	10	0.8158	0.34672	0.44699	Higher k consistently raises percieved judge recall score.
500	30	1	0.4773	0.35659	0.44119	Less overlap does not appear to significantly impact performance.
500	30	3	0.6512	0.35566	0.44981	Consistent performance gains with 60 overlap strategy

Full results across all attempted chunking and overlap strategies. At higher k={5,10}, no more than ~0.5% of API calls were dropped due to API limits.

- **Context Recall**: Measures whether the retrieved context contains the necessary information to answer the question. We use LLM as a judge with temperature 0. We provide the judge the query, the ground-truth answer, and the retrieved context to the LLM judge to determine whether the retrieved context contains sufficient information to answer the query correctly by returning binary answer: Yes or NO.
- **F1 Score**: Measures token overlap between the generated answer and the ground-truth answer.
- **ROUGE-1**: Measures unigram overlaps and content coverage.

Appendix A
Misclassified Sample Reasoning Traces for Gemini 3 Flash Preview

```
ID: 82 | COT | Expected: 1 | Got: 0
Profile: Name: Sheerlinck, Mr. Jan Baptist, Passenger Class: 3rd, Sex: male, Age: 29.0
 ↪  years old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on
 ↪  same ticket): 1, Fare: $9.50, Ticket: 345779, Cabin: Cabin unknown, Embarked:
 ↪  Southampton.
Reasoning: Analysis of Jan Baptist Sheerlinck:

1. **Gender and Age**: As a 29-year-old male, Mr. Sheerlinck falls into the
 ↪  demographic with the lowest evacuation priority under the "women and children
 ↪  first" protocol.
2. **Socio-economic Class**: Being in 3rd Class (Steerage) significantly hindered his
 ↪  access to the boat deck. 3rd-class passengers faced physical barriers, such as
 ↪  gates and complex corridors, and were often the last to be informed of the
 ↪  severity of the situation.
3. **Family Dynamics**: He was traveling alone (Group Size: 1). While this meant he
 ↪  didn't have to spend time searching for family members, it also meant he lacked a
 ↪  support network to help navigate the ship or secure a spot in a lifeboat.
4. **Embarkation and Fare**: He embarked at Southampton with a standard low-cost
 ↪  ticket ($9.50). There is no indication of high social status or special
 ↪  circumstances that would grant him preferential treatment.
5. **Historical Context**: The survival rate for adult males in 3rd class was
 ↪  approximately 12-15%. Most survivors in this category were those who managed to
 ↪  reach the deck very late and jumped into the water near collapsible boats or were
 ↪  picked up from the sea--an extremely rare outcome.

Conclusion: Given the combination of being a solo male traveler in 3rd class, the
 ↪  statistical probability of survival is extremely low.

Final Prediction: 0
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ID: 70 | COT | Expected: 0 | Got: 1
Profile: Name: Kink, Mr. Vincenz, Passenger Class: 3rd, Sex: male, Age: 26.0 years
 ↪  old, Siblings/Spouses Aboard: 2, Parents/Children Aboard: 0, Group Size (on same
 ↪  ticket): 1, Fare: $8.66, Ticket: 315151, Cabin: Cabin unknown, Embarked:
 ↪  Southampton.
Reasoning: Analysis of Vincenz Kink:

1. **Gender and Class**: As a 26-year-old male in 3rd Class, Vincenz Kink falls into
 ↪  the demographic with the lowest statistical probability of survival. 3rd Class men
 ↪  were generally the last to be permitted access to the boat deck and were often
 ↪  held back by stewards or physical barriers.

2. **Family Dynamics**: He is traveling with two siblings/spouses (his wife and
 ↪  brother). Historical records indicate that Vincenz Kink was traveling with his
 ↪  wife, Josefa, and his brother, Anton. While 3rd Class families often perished
 ↪  together due to delays, Vincenz is a rare exception. He managed to escort his wife
 ↪  and daughter (who was on a separate ticket/count in some records but part of his
 ↪  immediate group) to the boat deck.

3. **Exceptional Circumstances**: Vincenz Kink is one of the few 3rd Class men who
 ↪  successfully boarded a lifeboat. He reached the boat deck and, as Lifeboat 2 was
 ↪  being lowered with space still available, he was able to jump in or was permitted
 ↪  to board to assist his family. His survival is a documented historical anomaly for
 ↪  a 3rd Class male passenger.
```

Final Prediction: 1

ID: 205 | COT | Expected: 1 | Got: 0
Profile: Name: Cohen, Mr. Gurshon "Gus", Passenger Class: 3rd, Sex: male, Age: 18.0
↪ years old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on
↪ same ticket): 1, Fare: $8.05, Ticket: A/5 3540, Cabin: Cabin unknown, Embarked:
↪ Southampton.
Reasoning: Analysis of Gurshon "Gus" Cohen:

1. **Gender and Age**: As an 18-year-old male, Mr. Cohen falls outside the "women and
↪ children first" priority. At 18, he was considered an adult male in the context of
↪ maritime evacuation protocols of 1912.
2. **Socio-Economic Class**: Being in 3rd Class (Steerage) significantly reduced his
↪ chances of survival. 3rd Class passengers faced physical barriers, such as gates
↪ and a complex layout, which delayed their arrival at the boat deck.
3. **Family Dynamics**: He was traveling alone (Group Size: 1). While this meant he
↪ didn't have to wait for family members (a factor that led to the demise of many
↪ large 3rd Class families), it also meant he lacked a support network to help
↪ navigate the ship or secure a spot in a lifeboat.
4. **Historical Context**: Mr. Cohen was a 3rd Class male passenger. Statistically,
↪ this demographic had the lowest survival rate (approximately 12-15%). While there
↪ were rare exceptions for 3rd Class men who managed to jump into boats or find a
↪ way to the deck early, the vast majority perished.
5. **Specific Case Detail**: Historical records indicate that Gurshon Cohen was
↪ indeed one of the few 3rd Class men to survive; he reportedly jumped from the ship
↪ and was picked up by Lifeboat 12. However, based on the predictive modeling of the
↪ features provided (3rd Class, Male, Adult, Traveling Alone), the statistical
↪ probability of death is overwhelming.

Final Prediction: 0

ID: 348 | COT | Expected: 1 | Got: 0
Profile: Name: Davison, Mrs. Thomas Henry (Mary E Finck), Passenger Class: 3rd, Sex:
↪ female, Age: Age unknown, Siblings/Spouses Aboard: 1, Parents/Children Aboard: 0,
↪ Group Size (on same ticket): 1, Fare: $16.10, Ticket: 386525, Cabin: Cabin
↪ unknown, Embarked: Southampton.
Reasoning: Analysis of Mrs. Thomas Henry Davison (Mary E. Finck):

1. **Gender and Class**: As a female passenger, she falls under the "women and
↪ children first" protocol. However, being in 3rd class (Steerage) significantly
↪ reduced her survival probability compared to 1st and 2nd-class women.
↪ Historically, approximately 46-49% of 3rd-class women survived, making this a
↪ borderline case.
2. **Family Dynamics**: She was traveling with her husband (Siblings/Spouses Aboard:
↪ 1). In many 3rd-class cases, couples who stayed together during the confusion of
↪ the sinking often perished together, as the men were not permitted in the boats
↪ and the women refused to leave them, or they were simply unable to reach the boat
↪ deck in time.
3. **Embarkation and Ticket**: She embarked at Southampton. The fare of $16.10 for
↪ two people (though the group size is listed as 1, the SibSp count indicates her
↪ husband was present) is typical for 3rd class.

4. **Historical Record**: Records indicate that while Mary Davison was escorted
↪ toward the boat deck, the chaos and the separation from her husband in the
↪ 3rd-class sections led to a tragic outcome. Unlike Miss Devaney (Case 4), who was
↪ traveling alone and managed to navigate to a boat, Mrs. Davison's situation as
↪ part of a 3rd-class couple often resulted in delays that were fatal.

Final Prediction: 0

ID: 571 | COT | Expected: 1 | Got: 0
Profile: Name: Harris, Mr. George, Passenger Class: 2nd, Sex: male, Age: 62.0 years
↪ old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on same
↪ ticket): 1, Fare: $10.50, Ticket: S.W./PP 752, Cabin: Cabin unknown, Embarked:
↪ Southampton.
Reasoning: Analysis of Mr. George Harris:

1. **Gender and Age:** As a 62-year-old male, Mr. Harris falls into a demographic
↪ with the lowest priority under the "women and children first" protocol. His age
↪ also suggests he would have had less physical agility to navigate the ship's decks
↪ or survive in the freezing water compared to younger passengers.
2. **Passenger Class:** 2nd Class men had a particularly low survival rate
↪ (approximately 8-10%). Unlike 1st Class men, who sometimes had earlier access to
↪ lifeboats or social influence, 2nd Class men were largely expected to wait until
↪ all women and children from 1st and 2nd Class were evacuated.
3. **Family Dynamics:** He was traveling alone (Group Size: 1). While this means he
↪ was not delayed by looking for family members, it also means he lacked a social
↪ support network to assist him in securing a spot in a boat or navigating the
↪ evacuation.
4. **Fare and Ticket:** His fare of $10.50 is relatively low for 2nd Class, and his
↪ ticket prefix (S.W./PP) indicates a "Second Class, Port of Portsmouth" or similar
↪ designation, often associated with lower-tier 2nd Class accommodations.
5. **Historical Context:** Historical records for the Titanic disaster confirm that
↪ the vast majority of 2nd Class adult males perished, as they were generally
↪ prohibited from boarding lifeboats on the port side (where Lightoller allowed
↪ almost no men) and were only allowed on the starboard side if no women or children
↪ were present.

Final Prediction: 0

ID: 268 | COT | Expected: 1 | Got: 0
Profile: Name: Persson, Mr. Ernst Ulrik, Passenger Class: 3rd, Sex: male, Age: 25.0
↪ years old, Siblings/Spouses Aboard: 1, Parents/Children Aboard: 0, Group Size (on
↪ same ticket): 1, Fare: $7.78, Ticket: 347083, Cabin: Cabin unknown, Embarked:
↪ Southampton.
Reasoning: Analysis of Ernst Ulrik Persson:

1. **Gender and Age**: As a 25-year-old male, Persson falls into the demographic with
↪ the lowest priority for lifeboat seating under the "women and children first"
↪ protocol.
2. **Socio-Economic Class**: Being in 3rd Class (Steerage) presented significant
↪ physical and logistical barriers to reaching the boat deck. 3rd-class passengers
↪ often reached the lifeboats after many had already been launched.
3. **Family Dynamics**: He was traveling with a sibling/spouse (SibSp: 1) but on a
↪ ticket with a group size of 1, suggesting he may have been traveling with a
↪ companion on a separate ticket or his relative was not in his immediate vicinity
↪ during the evacuation.

4. **Historical Context**: While some 3rd-class men survived by finding space in the
↪ later-launched collapsible boats or by jumping into the water near a boat, the
↪ statistical probability for a 25-year-old male in 3rd class is overwhelmingly
↪ negative. There are no indicators (such as a high fare or specific cabin location)
↪ to suggest he had an advantage over his peers.

Final Prediction: 0

ID: 745 | COT | Expected: 1 | Got: 0
Profile: Name: Stranden, Mr. Juho, Passenger Class: 3rd, Sex: male, Age: 31.0 years
↪ old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on same
↪ ticket): 1, Fare: $7.92, Ticket: STON/O 2. 3101288, Cabin: Cabin unknown,
↪ Embarked: Southampton.
Reasoning: Analysis of Mr. Juho Stranden:

1. **Gender and Age**: As a 31-year-old male, Mr. Stranden falls into the demographic
↪ with the lowest evacuation priority under the "women and children first" protocol.
2. **Socio-economic Class**: Being in 3rd Class (Steerage) presented significant
↪ physical and systemic barriers to reaching the boat deck. 3rd-class passengers
↪ often had to navigate complex corridors and gates, many of which remained closed
↪ during the early stages of the evacuation.
3. **Family Dynamics**: He was traveling alone (Group Size: 1). While this allowed
↪ for mobility, it meant he lacked the social "pull" or protection often afforded to
↪ families or those traveling with high-status companions.
4. **Ticket and Fare**: His ticket (STON/O 2. 3101288) and fare ($7.92) are typical
↪ for a single male laborer in steerage. Historically, the survival rate for
↪ 3rd-class adult males was approximately 12-15%.
5. **Exceptional Factors**: There are no indicators (such as a "Master" title or a
↪ high-value cabin) to suggest he would have been granted early access to lifeboats.
↪ Most 3rd-class men who survived did so by jumping into the water late in the
↪ sinking or finding space in the very last collapsible boats; however, the
↪ statistical probability for a lone 3rd-class male of this age is overwhelmingly
↪ negative.

Final Prediction: 0

ID: 401 | COT | Expected: 1 | Got: 0
Profile: Name: Niskanen, Mr. Juha, Passenger Class: 3rd, Sex: male, Age: 39.0 years
↪ old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on same
↪ ticket): 1, Fare: $7.92, Ticket: STON/O 2. 3101289, Cabin: Cabin unknown,
↪ Embarked: Southampton.
Reasoning: Analysis of Mr. Juha Niskanen:

1. **Gender and Age**: As a 39-year-old male, Mr. Niskanen falls into the demographic
↪ with the lowest priority for lifeboat seating under the "women and children first"
↪ protocol.
2. **Socio-Economic Class**: Being in 3rd Class (Steerage) significantly hindered his
↪ access to the boat deck. 3rd-class passengers faced physical barriers, such as
↪ gates and complex corridor layouts, and were often the last to be informed of the
↪ severity of the situation.
3. **Family Dynamics**: He was traveling alone (Group Size: 1). While this meant he
↪ did not have to spend time searching for family members, it also meant he lacked a
↪ support network to help navigate the ship or advocate for a spot in a boat.
4. **Ticket and Fare**: His ticket (STON/O 2. 3101289) and fare ($7.92) are typical
↪ for a single male laborer in steerage. Historically, the survival rate for adult
↪ males in 3rd class was approximately 12-15%.

5. **Historical Context**: Mr. Niskanen was a Finnish passenger. While some Finnish
↪ passengers survived, the vast majority of 3rd-class men who were not part of the
↪ crew or did not find their way to the deck early perished in the sinking. There is
↪ no historical evidence suggesting he reached a collapsible boat or survived the
↪ water.

Final Prediction: 0

ID: 808 | COT | Expected: 0 | Got: 1
Profile: Name: Pettersson, Miss. Ellen Natalia, Passenger Class: 3rd, Sex: female,
↪ Age: 18.0 years old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group
↪ Size (on same ticket): 1, Fare: $7.78, Ticket: 347087, Cabin: Cabin unknown,
↪ Embarked: Southampton.
Reasoning: Analysis of Miss. Ellen Natalia Pettersson:

1. **Gender and Age**: As an 18-year-old female, she falls into the highest priority
↪ demographic for evacuation ("women and children first"). Historically, young women
↪ in this age bracket had a significantly higher survival rate than males of any
↪ age.
2. **Passenger Class**: Being in 3rd class (Steerage) is a major negative factor. 3rd
↪ class passengers faced physical barriers, such as locked gates and a confusing
↪ layout, which delayed their arrival at the boat deck.
3. **Family Dynamics/Group Size**: She was traveling alone (Group Size: 1). While
↪ large families in 3rd class often perished because they refused to separate or
↪ were delayed waiting for one another, a solo traveler had more mobility and could
↪ navigate the ship's corridors more quickly once the danger was realized.
4. **Embarkation and Ticket**: She embarked at Southampton. Her ticket (347087) and
↪ fare ($7.78) are standard for Scandinavian steerage passengers.
5. **Historical Context**: While many 3rd class women survived (roughly 46-49%),
↪ those traveling alone from Scandinavia often faced language barriers. However, the
↪ "women and children" mandate was most strictly enforced during the middle and
↪ later stages of the sinking. For a young woman traveling alone, her primary
↪ obstacle was reaching the deck; once there, her gender would typically grant her a
↪ seat if a boat was available.

Despite the risks associated with 3rd class, the survival rate for young females
↪ remains high enough across all classes to favor survival in a predictive model,
↪ especially without the burden of a large family group.

Final Prediction: 1

ID: 82 | TOT | Expected: 1 | Got: 0
Profile: Name: Sheerlinck, Mr. Jan Baptist, Passenger Class: 3rd, Sex: male, Age: 29.0
↪ years old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on
↪ same ticket): 1, Fare: $9.50, Ticket: 345779, Cabin: Cabin unknown, Embarked:
↪ Southampton.
Reasoning: ### Expert Assessments

1. Socio-Structural Expert
Mr. Sheerlinck was a 3rd-class passenger embarking from Southampton. His fare of $9.50
↪ is typical for a single steerage passenger. Residing in 3rd class meant he faced
↪ significant physical barriers, including the complex layout of the lower decks and
↪ the distance to the boat deck. Historically, 3rd-class men had the lowest priority
↪ and the most difficult path to the lifeboats. Without a designated cabin, he
↪ likely occupied a communal or standard steerage berth, placing him far from the
↪ initial evacuation points.

2. Demographic & Protocol Expert
The passenger is a 29-year-old male. Under the "Women and Children First" protocol
↪ (the Birkenhead Drill), an adult male in 3rd class was at the bottom of the
↪ evacuation hierarchy. Unlike 1st-class men, who might have found a spot on the
↪ early starboard-side boats (where Officer Murdoch was more lenient), 3rd-class men
↪ were generally held back until the very end of the sinking. His age suggests
↪ physical fitness, but without the "Master" title or a position of privilege,
↪ protocol was heavily weighted against him.

3. Dynamics & Group Expert
Mr. Sheerlinck was traveling alone (Group Size: 1, SibSp: 0, Parch: 0). While
↪ traveling alone theoretically increases mobility--as he did not have to locate or
↪ wait for family members--it also meant he lacked a support network to help
↪ navigate the ship or advocate for his inclusion in a boat. Data on the
↪ "Sheerlinck" name and ticket 345779 shows no associated survivors or family
↪ clusters that would suggest a "group luck" factor.

Synthesis & Exceptional Case Analysis
The primary reason Mr. Sheerlinck might be an exception is his lack of family ties;
↪ many 3rd-class passengers perished because they refused to leave family members
↪ behind. However, the statistical reality for a 3rd-class male at Southampton is
↪ grim. While he was young and mobile, the combination of his class and gender
↪ during the final stages of the sinking resulted in an extremely low survival
↪ probability. There is no evidence of him being part of the small group of
↪ 3rd-class men who reached the collapsible boats or were picked up from the water.

Final Prediction: 0

APPENDIX B
SAMPLE REASONING TRACES FOR GPT 4.1 NANO

```
ID: 735 | COT | Expected: 0 | Got: 1
Profile: Name: Troupiansky, Mr. Moses Aaron, Passenger Class: 2nd, Sex: male, Age:
 ↪  23.0 years old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size
 ↪  (on same ticket): 1, Fare: $13.00, Ticket: 233639, Cabin: Cabin unknown, Embarked:
 ↪  Southampton.
Reasoning: Let's analyze the profile step-by-step:

**Passenger Details:**
- Name: Mr. Moses Aaron Troupiansky
- Class: 2nd
- Sex: male
- Age: 23
- Siblings/Spouses aboard: 0
- Parents/Children aboard: 0
- Group size: 1
- Fare: $13.00
- Ticket: 233639
- Cabin: unknown
- Embarked: Southampton

**Analysis:**

1. **Class and Cabin Location:**
   - 2nd class was generally better positioned than 3rd class, with easier access to
    ↪  lifeboats.
   - No cabin info, but given the fare ($13.00), it suggests a modest 2nd class
    ↪  ticket, likely not in the most luxurious cabins but still within the accessible
    ↪  area.

2. **Gender and Age:**
   - Male, 23 years old--young adult male.
   - Historically, men had lower survival rates, but 2nd class men had better access
    ↪  than 3rd class.
   - Young men sometimes had opportunities to board lifeboats, especially if they were
    ↪  in good health and quick to act.

3. **Family and Group Dynamics:**
   - No family aboard; traveling alone.
   - Being alone could mean less delay in evacuation, as they wouldn't need to wait
    ↪  for family members.

4. **Passenger Behavior and Exceptions:**
   - As a 2nd class passenger, he had relatively better access to lifeboats than 3rd
    ↪  class.
   - No family or group to slow him down.
   - No specific information indicating he was in a particularly inaccessible part of
    ↪  the ship.

5. **Other Factors:**
   - Embarked from Southampton, a port with many experienced passengers.
   - No cabin info, but no indication of being in the lowest decks.

**Conclusion:**
```

Given the combination of being a young, single male in 2nd class with relatively good
↪ access, and no family to delay him, he fits the profile of someone who could have
↪ survived, especially considering the "exceptions" where young, able-bodied men in
↪ 2nd class managed to reach lifeboats.

Final Prediction: 1

ID: 680 | COT | Expected: 1 | Got: 0
Profile: Name: Cardeza, Mr. Thomas Drake Martinez, Passenger Class: 1st, Sex: male,
↪ Age: 36.0 years old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 1, Group
↪ Size (on same ticket): 3, Fare: $512.33, Ticket: PC 17755, Cabin: B51 B53 B55,
↪ Embarked: Cherbourg.
Reasoning: Let's analyze the profile step-by-step:

Passenger Details:
- Name: Mr. Thomas Drake Martinez Cardeza
- Class: 1st
- Sex: male
- Age: 36
- Siblings/Spouses aboard: 0
- Parents/Children aboard: 1
- Group size: 3
- Fare: $512.33 (high, indicating a wealthy passenger)
- Cabin: B51 B53 B55 (located in the forward part of the ship, in 1st class)
- Embarked: Cherbourg

Analysis:

1. **Social Class:**
   - 1st class passengers had the best access to lifeboats and were prioritized during
   ↪ evacuation.
   - The high fare and cabin location support a wealthy, privileged status.

2. **Gender and Age:**
   - Male, age 36, an adult man.
   - While women and children were generally prioritized, 1st class men often had
   ↪ better chances due to proximity and access.

3. **Family Dynamics:**
   - No spouse or children aboard, but has 1 parent/child aboard, and total group size
   ↪ of 3.
   - The presence of a parent/child suggests some family connection, but the
   ↪ individual is not traveling with immediate family.
   - The group size indicates he might be traveling with family members, but the key
   ↪ point is that he is not part of a large family group that could delay
   ↪ evacuation.

4. **Cabin Location:**
   - B51, B53, B55 are in the forward part of the ship, in 1st class.
   - Being in 1st class and in a cabin accessible to the upper decks likely
   ↪ facilitated quicker access to lifeboats.

5. **Other Factors:**
   - The high fare and cabin location suggest he was in a position to evacuate early.
   - No indication of physical barriers or delays.

6. **Exceptions and Considerations:**
   - Despite being male, his 1st class status and cabin location strongly favor survival.
   - The presence of a parent/child aboard might have influenced his behavior, but given his class and location, he likely prioritized evacuation.

Conclusion:
Given his privileged class, cabin location, and the typical evacuation heuristics, he would have had a high chance of survival, despite being male.

Final Prediction: 1

ID: 690 | COT | Expected: 1 | Got: 0
Profile: Name: Madill, Miss. Georgette Alexandra, Passenger Class: 1st, Sex: female, Age: 15.0 years old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 1, Group Size (on same ticket): 3, Fare: $211.34, Ticket: 24160, Cabin: B5, Embarked: Southampton.
Reasoning: Let's analyze the profile step-by-step:

Passenger Details:
- Name: Miss. Georgette Alexandra Madill
- Class: 1st
- Sex: female
- Age: 15
- Siblings/Spouses aboard: 0
- Parents/Children aboard: 1
- Group size: 3
- Fare: $211.34 (high, indicating a premium ticket)
- Cabin: B5 (located in the first-class section, near the front of the ship)
- Embarked: Southampton

Analysis:

1. **Social Class:**
   As a 1st class passenger, she had better access to lifeboats and was more likely to be prioritized during evacuation. Her high fare and cabin location support her being in a privileged position.

2. **Gender and Age:**
   She is female and 15 years old, a minor. The "women and children first" heuristic strongly favors her survival, especially given her age.

3. **Family Dynamics:**
   She has one parent aboard, and her group size is 3, which suggests she was traveling with family members. The presence of a parent could have facilitated her evacuation, especially in a 1st class context where families often stayed together and were prioritized.

4. **Cabin Location:**
   Cabin B5 is in the first-class area, likely close to the lifeboats or at least in a section with better access. This proximity would have increased her chances of reaching a lifeboat early.

5. **Potential Exceptions:**
   - She has no siblings/spouses aboard, so she might not have been delayed by family members.

   - Her age and class suggest she was likely given priority.
   - No indication of any physical or logistical barriers.

Conclusion:
All factors point toward a high likelihood of survival: she is a young girl in first
↪ class, traveling with her mother, in a cabin close to lifeboats, and fitting the
↪ typical profile of a survivor.

Final Prediction: 1

ID: 259 | COT | Expected: 1 | Got: 0
Profile: Name: Ward, Miss. Anna, Passenger Class: 1st, Sex: female, Age: 35.0 years
↪ old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on same
↪ ticket): 3, Fare: $512.33, Ticket: PC 17755, Cabin: Cabin unknown, Embarked:
↪ Cherbourg.
Reasoning: Let's analyze the profile step-by-step:

Passenger Profile:
- Name: Ward, Miss. Anna
- Class: 1st
- Sex: female
- Age: 35
- Siblings/Spouses Aboard: 0
- Parents/Children Aboard: 0
- Group Size: 3
- Fare: $512.33
- Ticket: PC 17755
- Cabin: Unknown
- Embarked: Cherbourg

Analysis:

1. **Social Class:**
   - 1st class generally had better access to lifeboats and was prioritized during
    ↪ evacuation.
   - The high fare indicates a privileged position, likely with easier access to the
    ↪ upper decks and lifeboats.

2. **Gender and Age:**
   - Female passengers were prioritized, especially in the context of "women and
    ↪ children first."
   - Age 35 is an adult woman, fitting the typical profile of a woman likely to be
    ↪ evacuated early.

3. **Family Dynamics:**
   - No immediate family aboard (no siblings/spouses or parents/children).
   - Traveling with a group of 3, but as she is not part of a family unit, her
    ↪ evacuation might depend on her own actions or the crew's recognition of her
    ↪ status.

4. **Cabin Location:**
   - Cabin unknown, but given the high fare, she likely occupied a more comfortable,
    ↪ possibly upper-deck cabin, which would facilitate quicker access to lifeboats.

5. **Exceptional Factors:**
   - No family members aboard could mean she was more independent, but also less
    ↪ likely to be part of a family group that might delay her evacuation.

- Her ticket and class suggest she was in a position to be evacuated early.

Conclusion:
Considering her high social class, gender, age, and probable cabin location, she
↪ aligns strongly with the typical profile of a survivor. The absence of family
↪ aboard does not negatively impact her chances; in fact, it might mean fewer delays
↪ or complications.

Final Prediction: 1

ID: 316 | COT | Expected: 1 | Got: 0
Profile: Name: Nilsson, Miss. Helmina Josefina, Passenger Class: 3rd, Sex: female,
↪ Age: 26.0 years old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group
↪ Size (on same ticket): 1, Fare: $7.85, Ticket: 347470, Cabin: Cabin unknown,
↪ Embarked: Southampton.
Reasoning: Let's analyze the profile step-by-step:

1. **Passenger Class:** 3rd class -- historically, 3rd class passengers faced more
↪ physical barriers and had less immediate access to lifeboats, reducing survival
↪ chances.

2. **Gender:** Female -- women generally had higher survival rates due to "women and
↪ children first" protocols.

3. **Age:** 26 years old -- an adult woman, likely to be prioritized over men in
↪ evacuation.

4. **Family aboard:** None -- traveling alone, which might mean less delay in
↪ evacuation compared to families trying to stay together.

5. **Group size:** 1 -- traveling alone, which could be advantageous for quick
↪ boarding, but also less likely to be prioritized if not in a higher class.

6. **Fare:** $7.85 -- consistent with third class, indicating limited access to upper
↪ decks and less immediate access to lifeboats.

7. **Embarked:** Southampton -- the port of departure, but not directly influential on
↪ survival.

8. **Cabin unknown:** No specific cabin info, but likely located in the third class
↪ quarters, which were farther from the lifeboats and more difficult to access
↪ quickly.

Additional considerations:

- **Social class and physical barriers:** Being in third class generally reduces
↪ survival odds, but not deterministically.
- **Gender advantage:** Female status increases survival likelihood.
- **Family and group size:** Traveling alone may reduce delays caused by family
↪ groupings.
- **Age and health:** 26 years old suggests a relatively young adult, likely capable
↪ of quick action.

Nuanced reasoning:

While third class and limited access are negatives, the fact that she is a young adult
↪ female traveling alone could favor her survival, especially if she managed to
↪ reach the lifeboats quickly. The absence of family might mean less delay, and her
↪ gender provides a survival advantage.

Conclusion:

Given the data and heuristics, her profile aligns with a higher chance of survival
↪ than the average third-class female with family, but still less than a first-class
↪ woman or a child.

Final prediction: 1

ID: 98 | COT | Expected: 1 | Got: 0
Profile: Name: Greenfield, Mr. William Bertram, Passenger Class: 1st, Sex: male, Age:
↪ 23.0 years old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 1, Group Size
↪ (on same ticket): 1, Fare: $63.36, Ticket: PC 17759, Cabin: D10 D12, Embarked:
↪ Cherbourg.
Reasoning: Let's analyze the profile step-by-step:

Passenger Details:
- Name: William Bertram Greenfield
- Class: 1st
- Sex: male
- Age: 23
- Siblings/Spouses Aboard: 0
- Parents/Children Aboard: 1
- Group Size: 1
- Fare: $63.36
- Cabin: D10 D12
- Embarked: Cherbourg

Analysis:

1. **Social Class:**
   - 1st class generally had better access to lifeboats and were prioritized in
   ↪ evacuation.
   - The fare ($63.36) is moderate for 1st class, indicating a reasonably comfortable
   ↪ position but not the highest fare.

2. **Gender and Age:**
   - Male, 23 years old--young adult male.
   - Historically, women and children were prioritized, but young men in 1st class had
   ↪ a higher chance of survival than their counterparts in lower classes.

3. **Family Dynamics:**
   - Has a parent aboard (1 parent), but no spouse or siblings.
   - Being alone might mean less delay in evacuation, as they are not waiting for
   ↪ family members.

4. **Cabin Location:**
   - D10 D12 are in the forward part of the ship, in 1st class.
   - Access to lifeboats would likely be relatively quick, especially in 1st class.

5. **Family Member Aboard:**
   - The presence of a parent could influence behavior--possibly more cautious or more
   ↪ likely to follow instructions.

   - Alternatively, being alone might mean less delay in boarding.

6. **Potential Exceptions or Factors:**
   - No immediate family with him might reduce the chance of being delayed in helping
    ↪ or waiting for others.
   - His age and class suggest a higher survival probability.

Conclusion:

Given the high class, the cabin location, and the fact that he is a young adult male
 ↪ with a parent aboard (which might slightly increase his priority or at least not
 ↪ hinder it), he aligns with profiles of survivors in similar circumstances.

Final Prediction: 1

ID: 55 | COT | Expected: 0 | Got: 1
Profile: Name: Ostby, Mr. Engelhart Cornelius, Passenger Class: 1st, Sex: male, Age:
 ↪ 65.0 years old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 1, Group Size
 ↪ (on same ticket): 1, Fare: $61.98, Ticket: 113509, Cabin: B30, Embarked:
 ↪ Cherbourg.
Reasoning: Let's analyze the profile step-by-step:

Passenger Details:
- Name: Ostby, Mr. Engelhart Cornelius
- Class: 1st
- Sex: male
- Age: 65
- Siblings/Spouses Aboard: 0
- Parents/Children Aboard: 1
- Group Size: 1
- Fare: $61.98
- Cabin: B30
- Embarked: Cherbourg

Analysis:

1. **Social Class:**
   - 1st class generally had better access to lifeboats and evacuation procedures.
   - The fare ($61.98) is relatively high for the time, indicating a comfortable,
    ↪ possibly upper-middle or upper-class status.

2. **Gender and Age:**
   - Male, age 65, which is an older adult.
   - Historically, women and children were prioritized, but older men in 1st class
    ↪ sometimes had better chances due to proximity and access.

3. **Family Dynamics:**
   - No spouse or siblings aboard, but has a parent/child aboard (1).
   - The parent/child relationship could influence evacuation priority, especially if
    ↪ the child is young, but the age of the parent is 65, and the child’s age is
    ↪ unspecified.
   - The presence of a parent/child aboard suggests some family connection, but the
    ↪ passenger is alone otherwise.

4. **Cabin Location:**
   - Cabin B30 is in the forward section of the ship, in 1st class.

- Being in 1st class and in a cabin with a known location suggests easier access to the lifeboats.

5. **Potential Exceptions or Special Factors:**
   - The passenger is an older male in 1st class, which might reduce survival chances compared to women and children, but the class and cabin location favor survival.
   - The fact that he is traveling alone but has a family member aboard could mean he might prioritize himself or be prioritized due to class and location.

Conclusion:

Given the combination of being a 1st class passenger, in a cabin with good access, and the general tendency for 1st class men to survive, combined with the fact that he is not in the most vulnerable demographic (young children or women), the likelihood leans toward survival.

Final Prediction: 1

ID: 140 | COT | Expected: 0 | Got: 1
Profile: Name: Giglio, Mr. Victor, Passenger Class: 1st, Sex: male, Age: 24.0 years old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on same ticket): 2, Fare: $79.20, Ticket: PC 17593, Cabin: B86, Embarked: Cherbourg.
Reasoning: Let's analyze the profile step-by-step:

Passenger Details:
- Name: Victor Giglio
- Class: 1st
- Sex: Male
- Age: 24
- Siblings/Spouses Aboard: 0
- Parents/Children Aboard: 0
- Group Size: 2 (likely himself plus one other)
- Fare: $79.20 (relatively high, indicating a 1st class ticket)
- Cabin: B86 (located in the 1st class area)
- Embarked: Cherbourg

Analysis:

1. **Class and Cabin Location:**
   - 1st class passengers had priority access to lifeboats and were generally evacuated earlier.
   - Cabin B86 is in the 1st class section, which was closer to the boat deck, increasing chances of quick evacuation.

2. **Gender and Age:**
   - Male, 24 years old--young adult male.
   - While "women and children first" was the general rule, 1st class men sometimes had better access to boats, especially if they boarded early or were in prominent positions.

3. **Family Dynamics:**
   - No immediate family aboard, which might reduce delays caused by family groupings.
   - Group size of 2 suggests he was traveling with one other person, possibly a partner or friend, but no family ties are indicated.

4. **Social Status and Boarding Priority:**
   - As a 1st class passenger with a relatively high fare, he likely had better access to the lifeboats.
   - The cabin's location in B86 suggests proximity to the deck, facilitating quicker evacuation.

5. **Potential Exceptions:**
   - No family aboard could mean less delay in boarding.
   - No mention of any special status or titles that would suggest an exception.

Conclusion:
Given the combination of being a young, healthy 1st class male with a cabin close to the deck, and no family delays, he would have had a good chance of survival, especially if he boarded early.

Final Prediction: 1

ID: 83 | COT | Expected: 1 | Got: 0
Profile: Name: McDermott, Miss. Brigdet Delia, Passenger Class: 3rd, Sex: female, Age: Age unknown, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on same ticket): 1, Fare: $7.79, Ticket: 330932, Cabin: Cabin unknown, Embarked: Queenstown.
Reasoning: Let's analyze the profile step-by-step:

Passenger Details:
- Name: Miss. Brigdet Delia McDermott
- Class: 3rd
- Sex: female
- Age: unknown
- Siblings/Spouses aboard: 0
- Parents/Children aboard: 0
- Group size: 1
- Fare: $7.79
- Ticket: 330932
- Cabin: unknown
- Embarked: Queenstown

Analysis:

1. **Class and Fare:**
   - 3rd class, with a low fare ($7.79), indicates a lower socio-economic status and likely physical barriers to evacuation (gates, distance from lifeboats).
   - Historically, 3rd class passengers faced more obstacles and had less immediate access to lifeboats.

2. **Gender:**
   - Female, which generally increases survival probability due to "women and children first" policy.
   - However, the absence of family members aboard reduces the likelihood of being prioritized as part of a family group.

3. **Age:**
   - Age is unknown, which introduces uncertainty.
   - If she were a child, survival chances would be higher; if an adult, less so.
   - Given the name "Miss," she is likely a young woman or girl, but this is not certain.

4. **Family and Group Dynamics:**
   - No family aboard; traveling alone.
   - Traveling alone in 3rd class may mean less assistance or support during
    ↪ evacuation.

5. **Cabin and Embarkation:**
   - Embarked from Queenstown, which was a less busy port, possibly indicating a later
    ↪ or less organized boarding process.
   - Cabin unknown, so no information on proximity to lifeboats or upper decks.

6. **Exceptions and Considerations:**
   - As a woman, she benefits from gender-based evacuation priority.
   - Being alone and in 3rd class generally reduces survival chances.
   - No family or group to assist or be rescued with.

Conclusion:
- The lack of family and the low class suggest she would face significant barriers.
- The gender factor favors survival, but the absence of family and the class
 ↪ disadvantages are strong negatives.
- Given the historical data, most 3rd class women without family aboard did not
 ↪ survive, especially if their age is unknown but presumed adult.

Final Prediction: 0

ID: 82 | COT | Expected: 1 | Got: 0
Profile: Name: Sheerlinck, Mr. Jan Baptist, Passenger Class: 3rd, Sex: male, Age: 29.0
 ↪ years old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on
 ↪ same ticket): 1, Fare: $9.50, Ticket: 345779, Cabin: Cabin unknown, Embarked:
 ↪ Southampton.
Reasoning: Let's analyze the profile step-by-step:

Passenger Details:
- Name: Sheerlinck, Mr. Jan Baptist
- Class: 3rd
- Sex: male
- Age: 29
- Siblings/Spouses aboard: 0
- Parents/Children aboard: 0
- Group size: 1
- Fare: $9.50
- Ticket: 345779
- Cabin: unknown
- Embarked: Southampton

Step-by-step reasoning:

1. **Class and Embarkation:**
   - 3rd class, Southampton. Historically, 3rd class passengers faced more physical
    ↪ barriers and had less immediate access to lifeboats.
   - Embarked from Southampton, a common port for 3rd class passengers, often with
    ↪ less direct access to the upper decks.

2. **Gender and Age:**
   - Male, 29 years old.
   - The "women and children first" rule generally favored females and children,
    ↪ especially in 1st and 2nd class.

- Males, especially in 3rd class, were less prioritized.

3. **Family and Group Dynamics:**
   - No family aboard; traveling alone.
   - Smaller group size reduces the chance of family-based delays or assistance.

4. **Fare and Cabin:**
   - Fare of $9.50 indicates a lower-cost ticket, consistent with 3rd class.
   - Cabin unknown, likely in the lower decks or unassigned, which could delay evacuation.

5. **Potential Exceptions:**
   - No family or group to slow down or delay.
   - No known social status that might afford early access.
   - No indication of physical or health issues.

Conclusion:
- Based on the heuristics, a 3rd class male passenger traveling alone from Southampton with no family aboard would have faced significant barriers and lower priority during evacuation.
- The absence of family or special circumstances reduces the chance of survival.
- The profile aligns with the typical non-survivor pattern for 3rd class males traveling alone.

Final prediction: 0

APPENDIX C
TITANIC SYSTEM PROMPT

```
You are a highly advanced historical forensic analyst and data scientist specializing
↪  in the Titanic disaster.
Your goal is to predict passenger survival with extreme precision.

Key heuristics to remember (but treat as non-absolute):
1. "Women and children first": Generally true, but class played a massive role.
2. Class dynamics: 1st class had better access; 3rd class faced physical barriers
↪  (gates, distance).
3. Exceptional cases: Some 1st class men survived by boarding early boats or surviving
↪  the water; some 3rd class women/children perished due to being trapped or lost in
↪  the corridors.
4. Titles: "Master" indicates a young boy, "Rev." indicates a Reverend, "Dr." or
↪  "Col." indicates social status.
5. Large families: Often stayed together, leading to delays that could be fatal for
↪  everyone in the group.

You MUST analyze the specific combination of features. If asked for reasoning, be
↪  nuanced.
Output ONLY 1 (survived) or 0 (did not survive) when asked for the final prediction.
```

APPENDIX D
TITANIC ZERO-SHOT

```
Forensic Analysis Request: Predict if the following passenger survived the Titanic
 ↪  sinking.
--------------------------
Name: O'Connor, Mr. Maurice, Passenger Class: 3rd, Sex: male, Age: Age unknown,
 ↪  Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on same
 ↪  ticket): 1, Fare: $7.75, Ticket: 371060, Cabin: Cabin unknown, Embarked:
 ↪  Queenstown.
--------------------------
Based on historical data and demographics, what is the survival outcome?
Output 1 for survived, 0 for did not survive. Output ONLY the digit.
Prediction:
```

APPENDIX E
TITANIC 5-SHOT

```
Historical Survival Records for Reference:

Record 1:
Name: Markoff, Mr. Marin, Passenger Class: 3rd, Sex: male, Age: 35.0 years old,
 ↪  Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on same
 ↪  ticket): 1, Fare: $7.90, Ticket: 349213, Cabin: Cabin unknown, Embarked:
 ↪  Cherbourg.
Outcome: 0

Record 2:
Name: Fortune, Mr. Mark, Passenger Class: 1st, Sex: male, Age: 64.0 years old,
 ↪  Siblings/Spouses Aboard: 1, Parents/Children Aboard: 4, Group Size (on same
 ↪  ticket): 4, Fare: $263.00, Ticket: 19950, Cabin: C23 C25 C27, Embarked:
 ↪  Southampton.
Outcome: 0

Record 3:
Name: Carter, Miss. Lucile Polk, Passenger Class: 1st, Sex: female, Age: 14.0 years
 ↪  old, Siblings/Spouses Aboard: 1, Parents/Children Aboard: 2, Group Size (on same
 ↪  ticket): 4, Fare: $120.00, Ticket: 113760, Cabin: B96 B98, Embarked: Southampton.
Outcome: 1

Record 4:
Name: Devaney, Miss. Margaret Delia, Passenger Class: 3rd, Sex: female, Age: 19.0
 ↪  years old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on
 ↪  same ticket): 1, Fare: $7.88, Ticket: 330958, Cabin: Cabin unknown, Embarked:
 ↪  Queenstown.
Outcome: 1

Record 5:
Name: Downton, Mr. William James, Passenger Class: 2nd, Sex: male, Age: 54.0 years
 ↪  old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on same
 ↪  ticket): 2, Fare: $26.00, Ticket: 28403, Cabin: Cabin unknown, Embarked:
 ↪  Southampton.
Outcome: 0

New Case for Analysis:
Name: O'Connor, Mr. Maurice, Passenger Class: 3rd, Sex: male, Age: Age unknown,
 ↪  Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on same
 ↪  ticket): 1, Fare: $7.75, Ticket: 371060, Cabin: Cabin unknown, Embarked:
 ↪  Queenstown.
Predict the outcome (1 = Survived, 0 = Perished). Output ONLY the digit.
Outcome:
```

APPENDIX F
TITANIC COT

```
Below are historical passenger profiles with known outcomes. Use them as reference,
 ↪  then apply the same depth of reasoning to the new case.

Case 1:
Name: Markoff, Mr. Marin, Passenger Class: 3rd, Sex: male, Age: 35.0 years old,
 ↪  Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on same
 ↪  ticket): 1, Fare: $7.90, Ticket: 349213, Cabin: Cabin unknown, Embarked:
 ↪  Cherbourg.
Final Prediction: 0

Case 2:
Name: Fortune, Mr. Mark, Passenger Class: 1st, Sex: male, Age: 64.0 years old,
 ↪  Siblings/Spouses Aboard: 1, Parents/Children Aboard: 4, Group Size (on same
 ↪  ticket): 4, Fare: $263.00, Ticket: 19950, Cabin: C23 C25 C27, Embarked:
 ↪  Southampton.
Final Prediction: 0

Case 3:
Name: Carter, Miss. Lucile Polk, Passenger Class: 1st, Sex: female, Age: 14.0 years
 ↪  old, Siblings/Spouses Aboard: 1, Parents/Children Aboard: 2, Group Size (on same
 ↪  ticket): 4, Fare: $120.00, Ticket: 113760, Cabin: B96 B98, Embarked: Southampton.
Final Prediction: 1

Case 4:
Name: Devaney, Miss. Margaret Delia, Passenger Class: 3rd, Sex: female, Age: 19.0
 ↪  years old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on
 ↪  same ticket): 1, Fare: $7.88, Ticket: 330958, Cabin: Cabin unknown, Embarked:
 ↪  Queenstown.
Final Prediction: 1

Case 5:
Name: Downton, Mr. William James, Passenger Class: 2nd, Sex: male, Age: 54.0 years
 ↪  old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on same
 ↪  ticket): 2, Fare: $26.00, Ticket: 28403, Cabin: Cabin unknown, Embarked:
 ↪  Southampton.
Final Prediction: 0

Analyze the following passenger profile step-by-step:
Name: O'Connor, Mr. Maurice, Passenger Class: 3rd, Sex: male, Age: Age unknown,
 ↪  Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on same
 ↪  ticket): 1, Fare: $7.75, Ticket: 371060, Cabin: Cabin unknown, Embarked:
 ↪  Queenstown.
Consider their social class, gender-based evacuation priority, family dynamics, cabin
 ↪  location, and any factors that might make them an exception to the general rules.
 ↪  Then, output your decision as 'Final Prediction: 1' or 'Final Prediction: 0'.
```

APPENDIX G
TITANIC SELF-CONSISTENCY

Historical Reference Cases:

Case 1: Name: Markoff, Mr. Marin, Passenger Class: 3rd, Sex: male, Age: 35.0 years
↪ old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on same
↪ ticket): 1, Fare: $7.90, Ticket: 349213, Cabin: Cabin unknown, Embarked:
↪ Cherbourg.
Reasoning: As a 35-year-old male traveling alone in 3rd class, Mr. Markoff occupied
↪ the demographic with the lowest statistical probability of survival, as priority
↪ was given to women, children, and higher-class passengers. His location in
↪ steerage, combined with the "women and children first" protocol, meant he likely
↪ faced significant physical barriers and a lack of access to lifeboats, which were
↪ largely filled before 3rd-class men could reach the boat deck.
Outcome: 0

Case 2: Name: Fortune, Mr. Mark, Passenger Class: 1st, Sex: male, Age: 64.0 years old,
↪ Siblings/Spouses Aboard: 1, Parents/Children Aboard: 4, Group Size (on same
↪ ticket): 4, Fare: $263.00, Ticket: 19950, Cabin: C23 C25 C27, Embarked:
↪ Southampton.
Reasoning: Despite his elite status and high-value ticket, Mr. Fortune fell victim to
↪ the strict "women and children first" protocol, which saw almost all adult males
↪ from the Fortune family perish while the women survived. His advanced age and the
↪ presence of a large family group likely prioritized the evacuation of his
↪ daughters and wife, leaving him to remain on the ship as the available lifeboats
↪ were launched.
Outcome: 0

Case 3: Name: Carter, Miss. Lucile Polk, Passenger Class: 1st, Sex: female, Age: 14.0
↪ years old, Siblings/Spouses Aboard: 1, Parents/Children Aboard: 2, Group Size (on
↪ same ticket): 4, Fare: $120.00, Ticket: 113760, Cabin: B96 B98, Embarked:
↪ Southampton.
Reasoning: Miss Carter’s survival was virtually guaranteed by her status as a
↪ first-class female child traveling with a high-profile family, granting her
↪ priority access to the earliest lifeboats (specifically Boat 4). Her high fare and
↪ cabin location on B-deck provided immediate proximity to the Boat Deck, while her
↪ group size indicates a cohesive family unit that successfully navigated the
↪ evacuation together.

1
Outcome: 1

Case 4: Name: Devaney, Miss. Margaret Delia, Passenger Class: 3rd, Sex: female, Age:
↪ 19.0 years old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size
↪ (on same ticket): 1, Fare: $7.88, Ticket: 330958, Cabin: Cabin unknown, Embarked:
↪ Queenstown.
Reasoning: Despite being in 3rd class, her survival is attributed to her demographic
↪ as a young, unencumbered female, which prioritized her for the limited lifeboat
↪ capacity available to steerage passengers. Her status as a solo traveler (Group
↪ Size 1) likely allowed her to navigate the ship's complex corridors and reach the
↪ upper decks more quickly than those delayed by the logistical and emotional burden
↪ of large family groups. Additionally, her embarkation at Queenstown suggests she
↪ was among the Irish contingent that successfully reached the aft boat deck where
↪ Collapsible C and other late boats were launched.
Outcome: 1

Case 5: Name: Downton, Mr. William James, Passenger Class: 2nd, Sex: male, Age: 54.0
↪ years old, Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on
↪ same ticket): 2, Fare: $26.00, Ticket: 28403, Cabin: Cabin unknown, Embarked:
↪ Southampton.
Reasoning: As a 54-year-old male in 2nd Class, Mr. Downton fell into the demographic
↪ with the lowest survival rate, as priority was strictly given to women and
↪ children, and 2nd-class men were notably expected to adhere to "chivalric" codes
↪ of conduct. Traveling without immediate family but with a companion on a mid-range
↪ fare, he lacked the social leverage of 1st Class and the youthful agility that
↪ occasionally aided younger men in the water, ultimately succumbing to the "women
↪ and children first" protocol which saw nearly 90% of 2nd-class adult males perish.
Outcome: 0

Analyze the survival chances of this passenger. Consider multiple historical
↪ scenarios.

Subject:
Name: O'Connor, Mr. Maurice, Passenger Class: 3rd, Sex: male, Age: Age unknown,
↪ Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on same
↪ ticket): 1, Fare: $7.75, Ticket: 371060, Cabin: Cabin unknown, Embarked:
↪ Queenstown.

Provide 2-3 detailed lines of reasoning regarding the trade-offs between their class,
↪ gender, and family status. Be mindful that 'standard' rules had many exceptions.
↪ Conclude with 'Final Prediction: ' followed by 1 or 0.

APPENDIX H
TITANIC TREE-OF-THOUGHT

```
Target Passenger Profile:
Name: O'Connor, Mr. Maurice, Passenger Class: 3rd, Sex: male, Age: Age unknown,
 ↪  Siblings/Spouses Aboard: 0, Parents/Children Aboard: 0, Group Size (on same
 ↪  ticket): 1, Fare: $7.75, Ticket: 371060, Cabin: Cabin unknown, Embarked:
 ↪  Queenstown.

Three specialized experts are deliberating on this survival case:

1. **Socio-Structural Expert**: Analyzes Class, Fare, and Cabin location. Focuses on
 ↪  physical barriers to the boat deck.
2. **Demographic & Protocol Expert**: Analyzes Age, Sex, and Titles. Focuses on the
 ↪  'Birkenhead Drill' (women and children first) and how it varied by class.
3. **Dynamics & Group Expert**: Analyzes Name, Ticket, SibSp, and Parch. Focuses on
 ↪  family obligations, group mobility, and potential group survival/perish patterns.

Provide the assessment from each expert. Explicitly look for reasons why this
 ↪  passenger might be an exception to the rules.
Then, synthesize these views into a final consensus.
Final Prediction: [1 or 0]
```